ZAMBIA UNIVERSITY COLLEGE OF TECHNOLOGY

ICT 2201 Database Systems

Final examination - Academic year 2024, semester 2

Time allowed: 3 hours Total marks: 100 Lecturer: Mr K. Mwape

Instructions: answer ALL questions. Write clearly. Show all working.

Question 1: ER modelling

(a) In an ER diagram, a weak entity is one that:

(b) A many-to-many relationship is implemented in a relational schema by:

(c) An attribute that can be calculated from other attributes is:

(d) Cardinality in an ER relationship describes:

Question 2: Normalisation

(a) A relation is in 1NF when:

(b) 2NF removes which kind of dependency?

(c) 3NF removes which kind of dependency?

(d) BCNF requires that every determinant is a:

Question 3: Transactions

(a) The A in ACID stands for:

(b) Which property ensures committed data survives a crash?

(c) A dirty read happens when a transaction reads data that is:

(d) Two transactions each waiting for a lock the other holds is called: